3-D Mapping using MATLAB

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Abstract*—This paper proposes a new technique for 3D surface imaging of objects using ultrasonic sensors combined with an Arduino Uno platform and further processing in MATLAB. The surface data obtained using ultrasonic technology will be further processed using a suite of advanced computational techniques to build a detailed digital representation of the object's surface. Stereographic projection, Num neighbor algorithms, Poisson's method, and point cloud generation are several important techniques used in the methodology. Stereographic projection is applied for mapping the acquired data into a suitable coordinate system such that the spatial relationships of the object are retained without any distortions. The spatial organization of the sensor data is managed by making use of the Num neighbor algorithm, optimizing the correspondence of the neighboring data points as well as enhancing the accuracy in the surface reconstruction. Poisson's method of surface reconstruction is applied to create a smooth continuous representation out of the sparse sensor data. Ultimately, point cloud generation is used to generate a comprehensive 3D model of the object. This is crucial for applications involving critical surface characterization. The proposed system has numerous advantages, such as relatively low-cost implementation and great adaptability. It is, therefore, a promising solution for an enormous range of applications, especially in the field of precise surface modeling of industries. It has an opportunity particularly to be applied vastly in mining territory. Through broad applications, such as terrain modeling, mine mapping, and spatial data analysis enhancement for mining operations, the system could allow for improved decision-making about mine planning, resource estimation, and safety analyses through providing an inexpensive source with accurate means of 3D surface reconstruction. The results of the project show that 3D surface imaging with ultrasonic sensors is possible and can be effective, thus providing a real alternative to the much more costly methods of laser scanning or photogrammetry. Moreover, the possibility of integrating sensors into Arduino as well as into the MATLAB environment for processing proves that the system has great potential for wide accessibility and real-time applications. This approach contributes not only to the improvement of 3D imaging but also provides a platform for further explorations and refinement in industrial and research contexts.

Keywords*— *Point Cloud, Stereographic projection, Num neighbor, Poisson's Method

INTRODUCTION

This paper talks about the integration of ultrasonic sensors ,Arduino uno and MATLAB simulation model to depict the real time surface imaging of any 3-D object . This project aims to build an ecosystem that will try to nullify human assistance in to explore in areas of danger . The surface mapping technique uses several mathematical concept such as conformal mapping which is used to map a function from one domain to the other, Stereographic projection used to open the surface of a sphere in the cartesian coordinate system . The ultrasonic sensor provides the raw data in the parameters of distance and angle which is then fed to Arduino. MATLAB software has an inbuilt Arduino ide package which then receives the raw data and begins to process The accurate imaging of the surfaces of objects in three-dimensional space plays an important role in many areas of industrial and research application, from terrain modelling to precision mapping and analysis of spatial data. One of the major drawbacks with more traditional methods such as laser scanning and photogrammetry is their substantial cost and complication requirements. This paper brings out an innovative, low-cost 3D surface-imaging using ultrasonic sensors combined with the Arduino Uno platform and further data processing performed in MATLAB. Ultrasonic sensors are selected for their flexibility as well as reduced costs while using the time-of-flight principle to acquire data about the surface. A sound wave emitted from the sensor bounces off the surface of the object and returns back to the sensor. It then employs a synergy of advanced computational methods in order to yield a high-fidelity 3D surface model from this information. The system which is being discussed in this

article makes use of a number of important techniques that guarantee coherence and continuity of the reconstructed surfaces. Stereographic projection transforms raw sensor data into a suitable coordinate space while keeping without distortion the spatial relations of the object. The spatial organization of sensor data is optimized using the Num neighbour algorithm. This would increase the accuracy of point correspondence between neighbours. Project applies Poisson's surface reconstruction method to skim data points that are mostly irregular in nature to give a smooth, free-state surface resulting from the ultrasonic sensors' captures. Furthermore, point cloud generation is used for creating a full 3D model of an object and it is applicable in applications where detailed surface characterization is required, such as in geospatial analysis, manufacturing, and engineering. This method is helpful mainly in cases where low-cost and adaptable solutions are necessary. The incorporation of ultrasonic sensors with the Arduino platform gives rise to a very adaptive and accessible solution, which is further enhanced by MATLAB's real-time data processing capabilities. In this regard, application prospects are wide and their potential application in the mining sector is particularly significant and lies mainly in terrain modelling, mine mapping, and enhancement of spatial data analysis, which might eventually lead to better decision-making processes with respect to estimation of resources, planning mines, and ensuring safety in mines. The results presented here more clearly demonstrate the possibility and effectiveness of this ultrasonic-based 3D imaging system as a valid alternative, offering a promising solution for those industries seeking to have accurate, cost-efficient surface modelling. . Ease of Use

LITERARTURE REVIEW

1)Data Aquisation through Ardiuno
work studied different digital embedded platforms which can be programmed using Arduino IDE software and which can be used for digital control and data acquisition applications [1].

2)Poisson's Point Cloud Artifacts and Reconstruction
provided a characterization of point clouds according to properties that have the most impact on reconstruction algorithms: sampling density, noise, outliers, misalignment, and missing data [3].

3)DC Servo Motor with Position Feedback System
control the DC servo motor, there are three different methods that can be used by using Arduino namely Arduino Integrated Drive Electronics (IDE), Support target for Simulink (Support Package) and Arduino Input/Output (IO) Package [2].

METHODLOGY

The prototype works on several principles such as point cloud regeneration , Poisson's artifacts , Num neighbour algorithm etc.

1) Data Aquisation using HC-SR04 Ultrasonic sensor
The sensors rotate for a predefined angle of pi/6 to give 3 readings per point. this raw data is then sent to Arduino uno for further data mapping. Two dc servo motor with specification **SG90** with torque equal to 1.2kg-cm, with a consumption of 3-7.2V, weighing 9gm and rotates 0.10sec/60*. Each servo serves a unique purpose, one for xy-plane and other for z-plane. An LED is also included in the hardware circuit which blinks/glows to signify that the mapping is in process. Power module supplies 5V to the servo motor since Arduino isn't sufficient to provide power to all the hardware components

2)Point Cloud Regeneration

The raw data from the HC-SR04 ultrasonic sensor is fed to MATLAB which uses point cloud library to map the surface of the object using the raw data by forming an array list of the data and also de noises the unwanted frequencies to provide a smooth image of the graph. Since only two parameters are known for the servo motor , position and angle thus the data is in spherical coordinate system which is the converted to Cartesian coordinate system using mathematical conversions

Equations

- $\mathbf{x} = \rho \sin\phi \cos\theta$
- $\mathbf{y} = \rho \sin\phi \sin\theta$
- $\mathbf{z} = \rho \cos\phi$

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} a_R \\ a_\theta \\ a_\phi \end{bmatrix}$$

Fig 1 – Shows Conversion from spherical to cartesian coordinate system

3)Implementation of Num Neighbour Algorithm and Poisson's Equation

Num neighbour algorithm attempts to nullify the overlapped input data from the HC-SR04 ultrasonic sensor using MATLAB. It de-noises the neighbouring unnecessary points to form a better , smooth and accurate graph of the surface of object. Implementation of mathematical concept Poisson's equation to map the surface of any 3-D object . A Poisson distribution is a discrete probability distribution. It gives the probability of an event happening a certain number of times (k) within a given interval of time or space. The Poisson distribution has only one parameter, λ (lambda), which is the mean number of events.

1C In plane polar coordinates, (r,θ), Poisson's equation for a potential $\varphi(r,\theta)$ generated by a source $\rho(r,\theta)$ is

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\varphi}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\varphi}{\partial\theta^2}=\rho\,.$$

Fig 2 – Depicts the mathematical formula for Poisson's equation

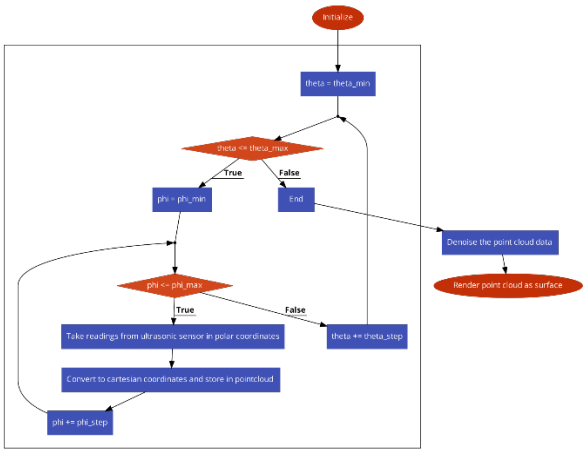


Fig 3 – Flowchart representation of the algorithm

Apparatus Required

Serial No	Component	Specification	Quantity
1)	Arduino Uno	Uno	1
2)	Power Module	5V, 800mA	1
3)	Ultrasonic senor	HC-SR04,40Hz,,15mA	1
4	Servo Motor	SG90,1.2Kg-cm,9gm,0.10sec/60*	2
5	Resistor	220 ohms	1
6	LED		1
7	Wire , Breadboard		1

Table 1- shows the tabulation for the components required for the hardware connection

Circuit Diagram

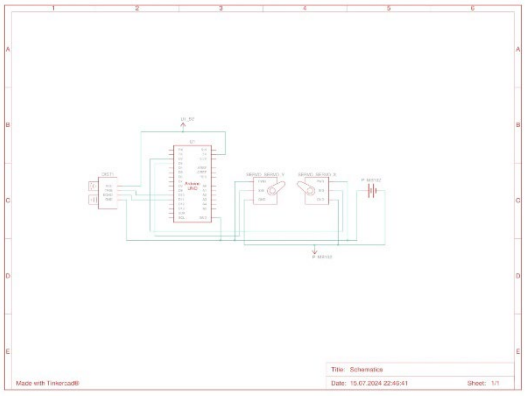


Fig 4 – shows the circuit diagram involved in the making of the project

Future Scope

The prototype can further be integrated with specialization concepts such as machine learning . The same would enhance the accuracy of the mapping by predicting the surface image and map the points accurately. Tensor Flow known for its wide range of libraries and flexibility will allow to integrate multiple concepts such as logistic , polyfit and Polynomial regression . The model will then have to be trained using these machine learning concepts to find better understanding of any 3-dimensional object and perform conformal mapping for the same in MATLAB

Result

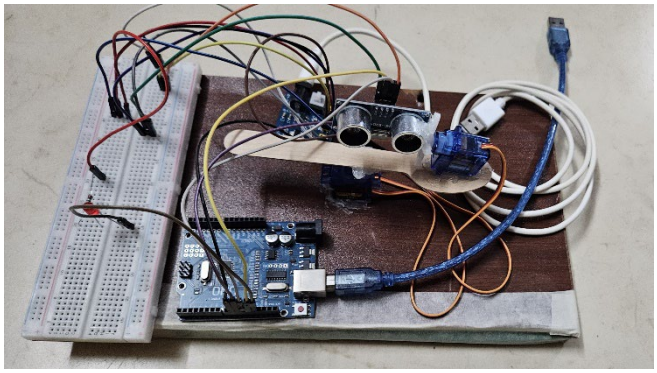


Fig 5 – shows the final hardware model of the project

Software Outputs

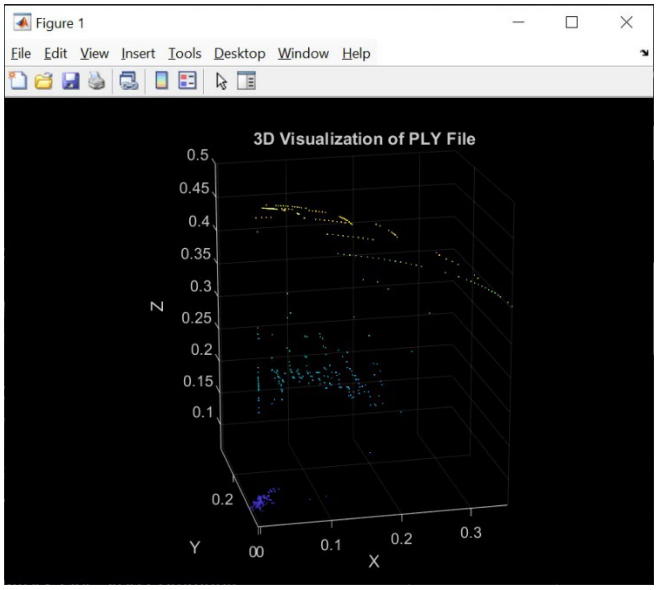


Fig 6 – shows the output pointcloud

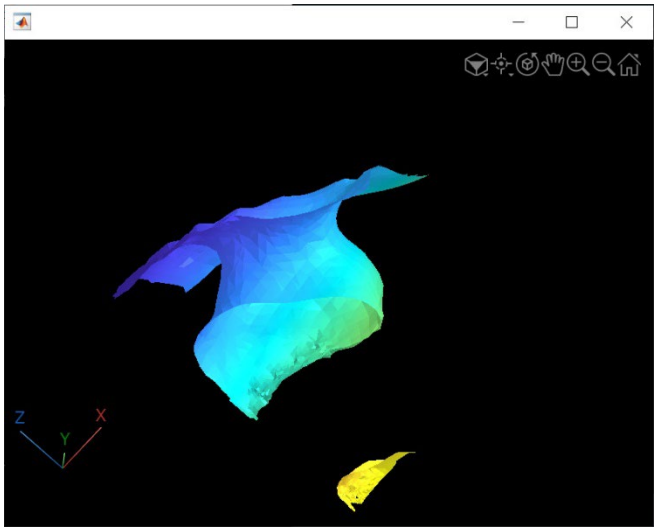


Fig 7 – shows the output reconstructed mesh

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The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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